

Validating a room-correction filter in context

Purpose

A correction filter should not be judged only from its own impulse response. The relevant endpoint is the loudspeaker, room and filter together. Phase rotation or stored energy in a filter may be correcting behavior already present in the loudspeaker-room system.

Filter-only tests remain useful: they catch sharp resonators, excessive group delay, long tails and export errors before deployment. They are screening tests, not a complete acoustic or perceptual verdict.

Required calculation

Use unsmoothed complex left and right measurements at every design position, the exact deployed filters, and a common acoustic timing reference. For each position p :

```
L_after(p,f) = L_before(p,f) * F_left(f)
R_after(p,f) = R_before(p,f) * F_right(f)

Stereo_before(p,f) = (L_before(p,f) + R_before(p,f)) / 2
Stereo_after(p,f)  = (L_after(p,f)  + R_after(p,f))  / 2
```

Apply filters before spatial averaging, while phase remains meaningful. A spatial RMS average has no coherent spatial phase and cannot reconstruct the stereo result at any seat.

Remove only common implementation latency. Retain the filter's relative phase, which is part of the correction under test. If exports omit DC or Nyquist, restrict analysis to measured frequencies and use identical tapered windows before and after. A rectangular frequency window creates a symmetric impulse that can be mistaken for pre-ringing.

What to report

Report tonal correction, group delay, precursor energy, late energy and gated-tone settling for left, right and coherent left-plus-right responses. Show medians and the full range across positions; do not hide a poor position inside one average.

Pre- and post-arrival energy have different perceptual meaning. Existing room decay after a transient does not automatically excuse new energy before it. Measure several precursor windows, such as earlier than 1, 5, 10 and 20 ms. A component confined close to the main impulse differs from a long pre-echo.

Band-limited bass impulses are intrinsically long. Their symmetric lobes must not be called filter pre-echo without comparison against the identical analysis window. Normalized late-energy fractions also change when EQ changes spectral weighting, so interpret them alongside level and main-arrival-referenced decay.

Role of `drc_acceptance.py`

`drc_acceptance.py` screens the filter for sharp features, bass group-delay excursions and gated-tone tails. Revision 2 uses full linear convolution, nine gate phases, a sustained quiet interval, late-echo observation, a pure-delay control and preserved censored trials. Regression tests cover rebounds, late echoes, phase coverage, censoring and zero-padding invariance.

Its thresholds are project build-quality heuristics. A pass does not establish audibility, acoustic improvement, spatial robustness or safe gain. Pre-peak energy is reported but does not enter pass/fail. The tail verdict uses the median of nine phases, and default tests stop at 200 Hz.

Evidence	Question answered
Filter-only acceptance	Is the coefficient set obviously pathological?
Multi-position contextual prediction	Does it improve the measured system, and at what timing cost?
Fresh corrected measurements	Did the real chain produce the prediction?
Level-matched listening	Is the resulting tradeoff preferable?

Manual correction and Dirac Live

Minimum-phase magnitude correction, crossover all-pass correction and inversion of position-dependent room excess phase are different operations. Crossover timing is a loudspeaker property that can remain consistent nearby. A reflection or cancellation that moves with the microphone is a poor candidate for exact phase inversion.

Professional mixed-phase systems face the same physics. Dirac says it limits pre-ringing by correcting excess phase only when it is common across measurement positions. Ordinary Dirac Live does not expose a user-controlled factorization equivalent to this project's common-bass filter plus separate left/right filters. Dirac Live Bass Control is related: its documentation describes a common low-frequency target, speaker-group-specific higher-frequency targets and cross-channel optimization. Its internal optimization is proprietary and is not the same inspectable construction.

The manual method has an advantage when kept auditable: the common and per-channel terms, their ranges, phase contribution and gain cost can be tested separately. Automation has another advantage: it can constrain a multi-position optimization consistently. The useful distinction is not homemade versus professional; it is whether stable behavior is separated from position-specific behavior and the combined response is verified.

References:

- [Dirac Live technical overview](#)
- [Dirac Live Bass Control filter design](#)
- [REW Group Delay graph](#)
- [DRC documentation](#)

Program-material headroom

Maximum frequency gain does not bound arbitrary music peaks after mixed-phase filtering. Phase rotation changes instantaneous summation, while resampling can reveal intersample peaks. Validate headroom with analytical checks, oversampled source peaks, offline convolution through the deployed path, demanding real recordings and a margin above the largest result.

A safety limit above 0 dBFS is an emergency abort threshold, not usable digital headroom. High-precision digital attenuation does not alter phase or create harmonic ringing; its practical cost is lower signal level relative to the downstream noise floor.

For each build retain measurement and coefficient hashes, acceptance output, per-position contextual results, coherent stereo results, stated precursor windows, real-program peak tests, attenuation, fresh corrected measurements when available, and the level-matched listening conclusion.